Automatic Cloud Detection in Remote Sensing Imagery Using Saliency-Based Mixed Features

Mi Wang, Xinsheng Wang, Yingdong Pi, and Shiyun Ke

***Abstract*— Cloud detection plays an important role in remote sensing image quality evaluation, information acquisition, and analysis. For most optical satellites, the acquired multispectral bands include only three visible bands and one near-infrared (NIR) band. How to achieve fast and accurate cloud detection with this limited number of bands is a problem worth studying. In this article, a cloud detection algorithm (SMFCD) based on a saliency-based mixed feature map of images is proposed. This algorithm first calculates the saliency characteristics from the band mean and haze-optimized transformation (HOT) map. The transmission map obtained from dark channel prior theory is used to combine the mixed feature map for subsequent detection. The proposed algorithm generates the initial thick cloud mask using a segmentation method based on the Otsu algorithm later. Guided filtering is then used to refine the cloud mask. The final cloud detection result is acquired through postprocessing. The algorithm is tested on six different datasets. It is shown that the algorithm can obtain good detection results for most images, with overall accuracies from 0.860 to 0.969 on these datasets. The average time consumption on these datasets reaches 4.4–42.1 s, and a fast version of the proposed algorithm reduces these times by about 50%. The proposed algorithm can be used for remote sensing image quality evaluation and subsequent application preprocessing with greater adaptability and flexibility than existing algorithms since it does not depend on specific satellite radiometric calibration coefficients.**

***Index Terms*— Cloud detection, guided filtering, Otsu algorithm, satellite images, visual saliency.**

I. Introduction

THE presence of large numbers of cloudy regions has a negative impact on the use of optical remote sensing images for object recognition and image classification. Worldwide, there are approximately 66% cloud cover over land surfaces [1], and therefore, cloud detection has become an important aspect of quality evaluation of remote sensing images and their applications. Early cloud detection algorithms focused mainly on threshold segmentation using multispectral physical characteristics, with development from fixed threshold to dynamic threshold methods [2], [3], [4], [5]. With the continuous expansion of the types of satellite sensors available, it is becoming difficult to meet the needs of cloud detection using such methods. As an alternative approach, the use of detection methods based on cloud characteristics has been proposed. Vásquez and Manian [6] computed second-order gray-level features, spectral features, and wavelet features to identify cloud pixels in MODIS images. Dong et al. [7] proposed a method in which superpixel segmentation is used to accelerate cloud detection in remote sensing images. Moreover, they applied the second angular moment derived from the gray level co-occurrence matrix (GLCM) [8] of each superpixel to eliminate cloud-like ground objects from cloud masks. Fisher [9] extracted geometric structural features to detect clouds in high-resolution SPOT images using morphological image processing. Li et al. [10] proposed an automatic multifeature combined (MFC) algorithm to detect cloud and cloud shadow in GaoFen-1 satellite imagery with high accuracy. Although these algorithms do a good job of cloud detection, there are still some problems with them. First, the input requirements for some algorithms are strict. For example, the input data of the MFC algorithm [10] consist of the top-of-atmosphere (TOA) reflectance of image bands whose value depends on radiometric calibration coefficients and meta-information such as the solar altitude angle. In addition, the radiometric calibration coefficients of a specific satellite may not be applicable to other satellites. The Fmask algorithm [4] takes seven Landsat TM/ETM+ spectral bands as input data, while the automatic cloud cover assessment (ACCA) algorithm [5] takes five of the eight Landsat ETM+ spectral bands. Second, some algorithms need to set a large number of parameters that are generally for a certain kind of specific satellite data. The MFC algorithm and the ACCA algorithm need to set 18 and 35 threshold parameters, respectively. All these parameters need to be retested and adjusted when applied to the different satellite images. Finally, some algorithms [7], [11], [12] can speed up the efficiency of cloud detection to a certain extent by superpixel segmentation of the original image, but it is difficult to effectively retain the detailed features of the boundary information of the cloud region.

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With the continuous developments in machine learning and artificial intelligence, methods, such as support vector machines (SVMs) and neural networks, are also being used as cloud detection classifiers [13], [14], [15], [16] and have acquired good results. In the last decade, convolutional neural networks have achieved a lot of achievements in the field of remote sensing image processing and have especially promoted the rapid development of remote sensing image cloud detection methods based on deep learning. Early DL-based cloud detection methods are designed to detect cloud on image blocks [17], [18], which depend on the quality of image block segmentation. When image blocks contain both cloud and noncloud pixels, it is easy to cause large detection errors. Later, pixel-based DL-based cloud detection algorithms emerged. For example, Wu and Shi [19] used the multilevel features extracted by CNNs to generate cloud probability maps and refined cloud masks through multiwindows guided filtering. Some scholars have proposed different cloud detection networks, such as CloudFCN [20] and RS-Net [21], based on the U-Net architecture [22] for image segmentation tasks. The MSCFF network [23] used multiscale convolutional feature fusion to improve the cloud detection accuracy of medium and high-resolution images. To improve the accuracy of cloud boundaries, Wang and Shi [24] proposed the ABNet to classify the cloud boundary points and further remedy the boundary information. He et al. [25] designed a lightweight network, termed DABNet, to reduce the high computational complexity, which further promoted the practical application of deep learning-based cloud detection on remote sensing images. Wu and Shi [26] fused scene information into cloud detection tasks and proposed the SAN model with robust on images of different scenes. Although DL-based cloud detection methods can achieve better accuracy, there are two inevitable drawbacks that need to be solved. The first one is that it is expensive and time-consuming to manually annotate pixelwise labels for massive remote sensing images. Most of the current DL-based cloud detection methods are designed for pixel-level classification based on supervised learning, which requires a lot of manual labeling information. Some works have proposed cloud detection models with no need of large manual labeled images based on weak supervision [27], [28], [29], but the research is relatively preliminary. Compared with the full supervision model, the accuracy of such methods is lower since errors may exist in the weakly supervised labels. In addition, different from the natural images, the shape and thickness of cloud vary greatly in remote sensing images. The thin clouds and cloud boundaries are often difficult to distinguish with the naked eye, which makes it difficult to manually label cloud pixels. The second problem is that DL-based cloud detection methods require the support of high-performance computing hardware such as GPU. For an application environment without GPU devices, the entire process is relatively time-consuming, whether it is processing by the CPU alone or transferring data to the GPU environment, which further limits the application of DL-based methods.

In the field of computer vision, saliency detection is usually used to extract the most prominent area in an image, especially for the common visible images [30]. Through saliency analysis, human observers can quickly locate and obtain the key information in an image. For remote sensing images, when there are clouds in an image, their positions can always be identified first by naked eye observation. The essence of this process is the detection of the significant features (i.e., clouds) in an image by human vision. Through saliency detection, the foreground and background of the image content can be segmented well, that is, the binary classification task of the image can be realized. Cloud detection in remote sensing images is also an image binarization task, in which cloud pixels are taken as the foreground and noncloud pixels are considered as the background.

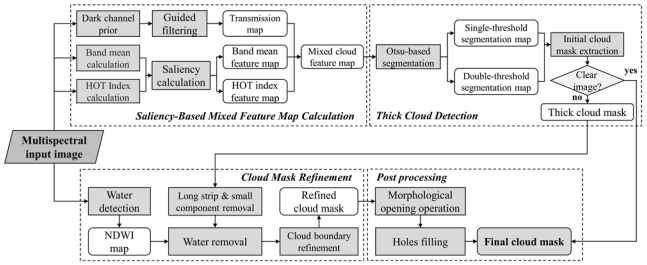


Fig. 1. Flowchart of the proposed SMFCD algorithm.

In this article, a saliency-based mixed feature cloud detection (SMFCD) algorithm is proposed for remote sensing imagery. This algorithm uses saliency calculation to construct a mixed cloud feature map and uses a modified Otsu segmentation to generate the thick cloud mask. This mask is then refined with guided filtering, and the final cloud mask is acquired after postprocessing. The proposed algorithm only takes the original digital number (DN) values of four bands [visible bands and the near-infrared (NIR) band]. Therefore, it does not depend on the specific satellite radiometric calibration coefficient to calculate the TOA reflectance, which makes it more universal and flexible than previous methods.

II. Method

A flowchart of our proposed SMFCD algorithm is shown in Fig. 1. The input data of the algorithm are the DN values of the blue, green, red (BGR), and NIR bands of satellite imagery. The algorithm first calculates the haze-optimized transformation (HOT) map, band mean map, and transmittance map of the original image. Then, it calculates the saliency map of the first two maps. The mixed cloud feature map is formed with two saliency maps and transmittance map. Image segmentation based on the Otsu algorithm is then carried out on the mixed cloud feature map, and this is followed by thick cloud detection. Refined processing techniques, such as geometric feature removal, water removal, and cloud boundary extension, are then applied to the thick cloud mask. The final cloud mask is obtained after some postprocessing operations.

A. Saliency-Based Mixed Feature Map Calculation

1) HOT Index Feature Calculation: The HOT index [31] has been widely used in cloud detection in Landsat images. By defining a clear line between these two different types of

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pixels to be drawn in the image spectral space of the blue and red bands, cloud pixels have larger HOT index values than noncloud pixels for most scenes. The HOT index is defined as follows:

$$\text{HOT} = \text{Band}_{\text{blue}} \sin\theta - \text{Band}_{\text{red}} \cos\theta \tag{1}$$

where $\text{Band}_{\text{Blue}}$ and Band_{red} are the DN values of the blue and red bands, respectively, and θ is the slope angle of clear line in spectral space. In this article, the HOT index is simplified as

$$\text{HOT} = \text{Band}_{\text{blue}} - 0.5\ \text{Band}_{\text{red}}. \tag{2}$$

2) Image Transmission Estimation:

a) Dark channel prior theory: Dark channel prior theory [32] is used in haze removal tasks for single RGB images. Considering that thin cloud in remote sensing images has similar features to haze in ordinary RGB images, some researchers have used this theory to remove thin cloud in remote sensing images and thereby improve the visual effect. In our proposed algorithm, we also introduce dark channel prior theory to improve the detection ability for thick clouds and suppress the features of noncloud pixels. Dark channel prior theory is based on a large number of statistical rules. By counting a large number of RGB images, He et al. [32] found that there are always some very dark pixels in the local regions of clear sky images. However, when an image contains haze (as well as thin cloud), the dark regions in the image will be affected and become gray or even white. Based on this, the following scattering model for hazed images is proposed:

$$\boldsymbol{I}(\boldsymbol{x}) = \boldsymbol{J}(\boldsymbol{x})\boldsymbol{t}(\boldsymbol{x}) + \boldsymbol{A}[1 - \boldsymbol{t}(\boldsymbol{x})] \tag{3}$$

where $\boldsymbol{x}$ is a pixel in the image, $\boldsymbol{I}(\boldsymbol{x})$ refers to the obtained image intensity, $\boldsymbol{J}(\boldsymbol{x})$ is the scene image intensity (also known as the clear image), $\boldsymbol{t}(\boldsymbol{x})$ is the transmission, which is between 0 and 1.0, and A is the atmospheric light intensity. The dark channel can be defined as follows:

$$\boldsymbol{J}^{\text{dark}}(\boldsymbol{x}) = \min_{\boldsymbol{y}\in\Omega(x)} \left(\min_{c\in\{r,g,b\}} \boldsymbol{J}^c(\boldsymbol{y}) \right) \tag{4}$$

where $\boldsymbol{J}^c, c \in \{r, g, b\}$, are the three channels of the RGB image and $\Omega(\boldsymbol{x})$ is a local window centered at $\boldsymbol{x}$.

According to the dark channel prior theory, J^{dark} should be close to zero. Given a fixed atmosphere light intensity A, the transmission can be computed as follows:

$$\tilde{t}(\boldsymbol{x}) = 1 - \omega \min_{\boldsymbol{y}\in\Omega(x)} \left(\min_{c} \frac{I^c(\boldsymbol{y})}{A^c} \right) \tag{5}$$

where ω is an adjustment factor that is used to retain some fog to make the image more realistic in terms of visual effects.

b) Guided filtering: Although the transmission map can be calculated through (5), the initial result is very rough due to the application of the local minimum in the calculation, and there is an inevitable "box effect" on the initial transmission map. To obtain finer results, a guided filter [33] is used for enhancement of image details. This is an edge-preserving filter, involving a guidance image and an input image. Based on the assumption that the output image is a local linear transform of the guidance image, the guided filter can be expressed mathematically as follows:

$$q_i = a_k G_i + b_k \quad \forall i \in \omega_k \tag{6}$$

where a_k and b_k are linear coefficients in a local window ω_k, i refers to a pixel in ω_k, and q_i and G_i are image DN values of pixel i in the output and guidance image, respectively. The model in (6) shows that the gradient of the output q is linearly related to the gradient of the guidance image G, and a_k and b_k can be solved by minimizing the noise $n$between the input image and the output image

$$n = \|\boldsymbol{q} - \boldsymbol{p}\| = \sum_{i\in\omega_k} \left[(a_k G_i + b_k - p_i)^2 + \varepsilon a_k^2 \right] \tag{7}$$

where $\boldsymbol{p}$ is the input image and ε is a regularization term. The solution is then obtained as follows:

$$a_k = \frac{\frac{1}{|\omega|}\sum_{i\in\omega_k} G_i p_i - \bar{p}_k \mu_k}{\delta_k^2 + \varepsilon} \tag{8}$$

$$b_k = \bar{p}_k - a_k \mu_k \tag{9}$$

where μ_k and δ_k^2 are the mean and variance of $\boldsymbol{G}$ in the window ω_k, respectively, $|\omega|$ is the number of pixels in ω_k, and $\bar{p}_k$ is the mean of $\boldsymbol{p}$ in ω_k.

Since pixel i will be covered by more than one window, the final output image q_i is computed by using the means of the coefficients of a and b from multiple windows. Thus, the result is given by

$$q_i = \frac{1}{|\omega|} \sum_{k\in\omega_i} a_k G_i + \frac{1}{|\omega|} \sum_{k\in\omega_i} b_k. \tag{10}$$

With the guided filter, the transmission map can be refined by using the original RGB image as the guidance image, and the output image will then preserve much greater details at the edges of objects. Given the efficiency of the original guided filter, a further development, namely, the fast guided filter, has been proposed [34]. With this filter, the input image and guidance image are downsampled before the calculation of coefficient maps a and b. These coefficient maps are then upsampled to the full resolution, and a linear operation is carried out on the original guidance image to obtain the final results. The computation of the guided filter can thereby be reduced from $O(N)$ complexity to $O(N/s^2)$, where s is the subsampling ratio.

3) LC Saliency Calculation: According to most psychological studies, human visual system is sensitive to motion contrast for image sequence and visual signal contrast, especially the color, intensity, and texture features for still images. Inspired by the latter capabilities of human vision, the spatial attention model adopted in Zhai and Shah's algorithm [35] is a fast and comprehensive saliency detection technique that uses color histograms only. The luminance contrast (LC) saliency value of a pixel x in an image I is described as follows:

$$\text{Sal}(x) = \sum_{\forall k\in I} \|I_x - I_k\| \tag{11}$$

where I_x and I_k are the DN values of pixels x and k, respectively, and $\|\cdot\|$ represents the distance between two DN values. For a given input image, the DN value I_x of each pixel

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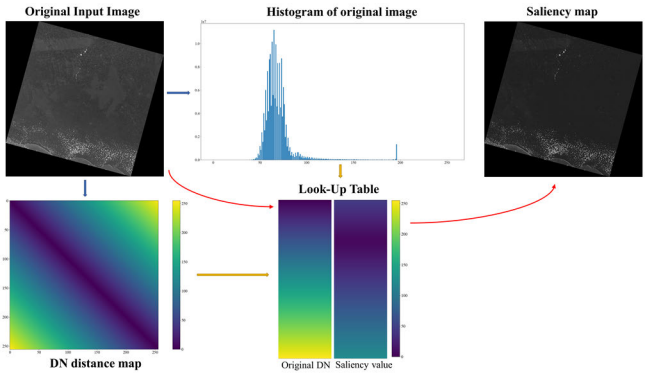


Fig. 2. Diagram of saliency calculation.

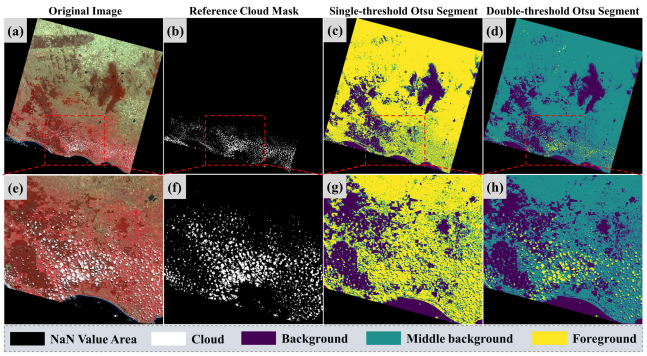


Fig. 3. Comparison of single- and double-threshold Otsu segmentation algorithms. (a) Original image (green, red, and near-infrared band combination). (b) Reference cloud mask. (c) Single-threshold Otsu segmentation result. (d) Double-threshold Otsu segmentation result. (e)–(h) Details of subfigure (a)–(d).

and the overall DN range are fixed. Therefore, the saliency calculation of each pixel can be further restricted to each DN value since terms with the same DN value in (11) can be merged, and the equation can be rewritten as follows:

$$\mathrm{Sal}(m) = \sum_{n=0}^{\mathrm{DN}_{\max}} f_n \|m - n\| \tag{12}$$

where m is the DN value of the current pixel, f_n is the frequency of DN value n, and $\mathrm{DN}_{\max}$ is the maximum DN value of the image. The frequencies of the DN value are expressed in the histograms, which are counted before saliency calculation. Since $n \in [0, \mathrm{DN}_{\max}]$, the distance between m and n is bounded to the same range and can be calculated as a symmetric map $D(m, n) = \|I_m \text{ t } I_n\|$ in advance. With the fixed map $D(\cdot, \cdot)$ and frequency $f(\cdot)$, the saliency value of any DN value p can be computed as $S(p) = \sum_{q=0}^{\mathrm{DN}_{\max}} f_q D(p, q)$. Therefore, the correspondence between the original DN values and the saliency values can be constructed as a lookup table, and the image significance calculation can be directly and efficiently realized by the pixel-by-pixel lookup table replacement without large amount of computation. The LC saliency calculation process is shown in Fig. 2.

4) Mixed Cloud Feature Map Generation: Although feature maps, such as the HOT index, can be useful in cloud detection in some images, it is often difficult to use a single feature to achieve better accuracy for some complex scenes. Therefore, this article proposes a mixed cloud feature map, which further enhances the difference between cloud pixels and noncloud pixels by calculating the saliency of the HOT map and the mean map of the BGR bands. At the same time, considering the existence of thin clouds, we use the transmittance map as an adjustment coefficient to improve the distinction between areas with and without cloud cover. The proposed mixed cloud feature map is represented as follows:

$$\mathrm{MCFM} = [1 - t(x)](0.5\ \mathrm{HOT}_{\mathrm{sal}} + 0.5\ \mathrm{MeanBGR}_{\mathrm{sal}}) \tag{13}$$

where $\mathrm{HOT}_{\mathrm{sal}}$ and $\mathrm{MeanBGR}_{\mathrm{sal}}$ are the saliency maps of the HOT map and the mean of the BGR bands, respectively.

B. Thick Cloud Detection

1) Segmentation Based on Otsu Algorithm: The Otsu algorithm [36] can be used to automatically select a threshold value for binarization. This algorithm is simple to calculate and is not affected by image brightness and contrast. Based on the idea of clustering, the Otsu algorithm divides an image into two parts according to the gray level such that the discriminant criterion measure between the two parts is maximum. The discriminant criterion measure is defined as follows:

$$g_k = \omega_0(\mu_0 - \mu_T)^2 + \omega_1(\mu_1 - \mu_T)^2 \tag{14}$$

where g_k is the between-class variance when the selected threshold is k, ω_0 and ω_1 are the pixel ratios of foreground and background, respectively, and μ_0, μ_1, and μ_T are the mean values of the foreground, background, and total images, respectively.

The best threshold is selected through calculation of the variance to find a suitable DN value at which to divide. However, this algorithm is not suitable in complex situations, such as cloud detection in remote sensing images [see, Fig. 3(c)]: in the cloud image segmentation task, most noncloud pixels, such as ground, high light buildings, and highly reflective water surfaces, cannot be distinguished from cloud pixels. We extend the original single-threshold Otsu algorithm to a double-threshold Otsu algorithm to provide better segmentation of cloud pixels. The discriminant criterion measure is modified as follows:

$$g'_{k1,k2} = \omega_0(\mu_0 - \mu_T)^2 + \omega_1(\mu_1 - \mu_T)^2 + \omega_2(\mu_2 - \mu_T)^2 \tag{15}$$

where $g'_{k1,k2}$ is the between-class variance when the selected double thresholds are k_1 and k_2, ω_0, ω_2, and ω_3 are the pixel ratios of three segmented regions, and μ_0, μ_1, and μ_2 are the corresponding mean values of these segmented regions. A comparison of the segmentation results obtained with the single- and double-threshold versions of the Otsu algorithm is shown in Fig. 3.

2) Initial Cloud Mask Extraction: For most images with obvious color contrast between cloud and noncloud objects, the single-threshold Otsu algorithm is able to perform segmentation and obtain a good initial mask. For images with highlighted objects or little color difference between clouds and noncloud objects, the double-threshold Otsu algorithm can accurately extract a rough cloud mask to ensure the accuracy

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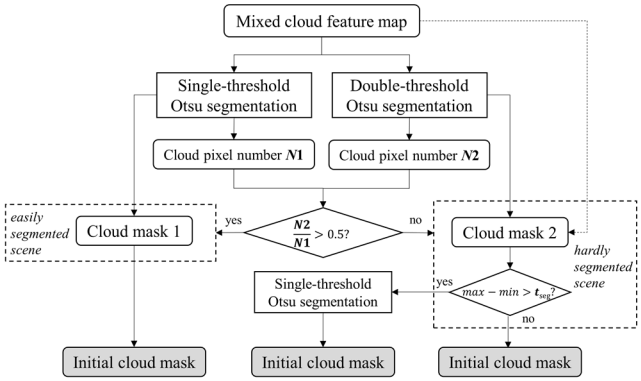


Fig. 4. Flowchart of initial cloud mask generation.

of cloud detection. However, since the double-threshold Otsu algorithm reduces the selection rate of cloud pixels expectedly, if this algorithm is used uniformly in all scenes, the omission rate of the initial cloud mask will be high. Therefore, we propose a strategy in which the different Otsu segmentation methods are used for different scenes. A flowchart of the procedure is shown in Fig. 4. We calculate the ratio between two numbers $\boldsymbol{N1}$ and $\boldsymbol{N2}$, where $\boldsymbol{N1}$ refers to the number of cloud pixels that appear in both of the segmentation results, and $\boldsymbol{N2}$ refers to the number of cloud pixels that appear in only the segmentation result from the double-threshold Otsu algorithm. In other words, the ratio between $\boldsymbol{N1}$ and $\boldsymbol{N2}$ means the proportion of intersecting cloud pixels in both segmentation results compared with the cloud pixels in the double-threshold Otsu segmentation result.

On the basis of this ratio, image scenes can be considered as either "easily segmented" or "hardly segmented." For the former scenes, we use single-threshold segmentation to ensure that more cloud pixels are extracted, whereas for the latter, double-threshold segmentation is adopted to ensure that fewer noncloud pixels are extracted by mistake. Through this strategy, single- or double-threshold Otsu segmentation results are both obtained, which is appropriate to ensure the accuracy of the initial cloud mask. For the latter scenes, further segmentation may be needed. If a big difference between the maximum and minimum of the mixed features occurs (e.g., greater than the threshold $\boldsymbol{t}_{\text{seg}}$), the double-threshold segmented result will be considered as highly misidentified, which requires a single-threshold Otsu segmentation to reduce its high commission errors.

3) Clear Image Judgment: After the initial mask has been obtained, it is necessary to judge whether it contains clouds to avoid misclassification of clear images. Considering that the initial cloud mask extracted at present is mainly concentrated in the core thick cloud area, its brightness should be high enough. Based on this assumption, the mean DN value of the BGR bands corresponding to the initial cloud mask pixel is calculated. If it is lower than a certain threshold $\boldsymbol{t}_{\text{clear}}$ (e.g., 150), the image is considered to be a clear image, and no subsequent operations need to be carried out.

C. Cloud Mask Refinement

1) Long Strip and Small Component Removal: The geometric features of noncloud objects, such as coastlines and roads, often have long strip shapes, whereas most cloud objects rarely have such shapes. Therefore, using the height-to-width ratio (HWR) of the minimum enclosing rectangle of each object, long strip-like objects can be easily detected and removed. Through the connected component labeling algorithm, the individual cloud components of the acquired cloud mask can be labeled so that the geometric features of each component can be calculated. The HWR can be expressed as follows:

$$\text{HWR} = \frac{\max(\text{height}, \text{width})}{\min(\text{height}, \text{width})}. \tag{16}$$

Considering that clouds sometimes appear as long strips, it is necessary to take the area of the object in addition to the value of HWR into account. Therefore, we set up a method to eliminate the long strip objects with the double constraint. First, the connected domain analysis of the cloud mask is carried out, and then, the area and HWR value of each labeled object are calculated, denoted as area and hwr, respectively. Then, for a small area object, if area $> \boldsymbol{t}_{\text{area1}}$ and hwr $< \boldsymbol{t}_{\text{hwr1}}$, the object is considered as noncloud. For a large-area object, if area $> \boldsymbol{t}_{\text{area2}}$ and hwr $< \boldsymbol{t}_{\text{hwr2}}$, the object is also excluded.

2) Water Removal: The normalized difference water index is given by

$$\text{NDWI} = \frac{\text{Band}_{\text{green}} - \text{Band}_{\text{nir}}}{\text{Band}_{\text{green}} + \text{Band}_{\text{nir}}} \tag{17}$$

where $\text{Band}_{\text{green}}$ and Band_{nir} are the DN values of the green and NIR bands, respectively. Fortunately, thick cloud and thin cloud pixels were labeled in the reference masks of the Landsat7 dataset [37]. Hence, we conducted statistics on the histogram of NDWI values at thick cloud and thin cloud pixels in the Landsat7 dataset, as shown in Fig. 5. We found experimentally that for images containing thin clouds, the NDWI histogram of cloud pixels shows a peak near 0.2, but it is generally less than 0.4. For other scenes, most values of NDWI histogram only have a peak near 0.0 but are generally less than 0.2. Therefore, we adopt double-threshold elimination when performing water removal. If more than half of the current cloud pixels have NDWI values greater than 0.2, a smaller threshold $\boldsymbol{t}_{\text{ndwi1}}$ (e.g., 0.2) is selected; otherwise, a larger threshold $\boldsymbol{t}_{\text{ndwi2}}$ (e.g., 0.4) is selected to ensure that most thin clouds will not be eliminated.

3) Cloud Boundary Refinement: The edge of a cloud cluster is usually dominated by thin clouds, and the transmittance and color between thin clouds and thick clouds are very distinguishable and change with the transformation of the ground objects under the cloud surface. Consequently, these cloud pixels are often missing in the thick cloud mask extracted in the early stage. Such pixels tend to be concentrated around thick clouds, but it is difficult to extract relatively complete and accurate cloud pixels by simply extending thick clouds outward. Due to the excellent boundary preserving properties of the guided filter [33], we use it with a large radius $\boldsymbol{t}_{\text{GF_}r}$ (e.g., 200) for cloud boundary refinement. The original BGR color image is used as the guidance image and the thick cloud mask as the input image. In the filtering results, those pixels with gray value greater than a certain threshold $\boldsymbol{t}_{\text{GF_}t}$ (e.g., 30) are considered as cloud pixels. The results of this cloud boundary refinement procedure are shown in Fig. 6.

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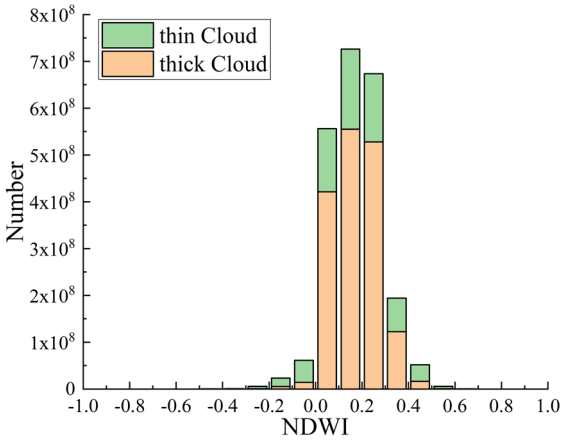


Fig. 5. NDWI histogram of thick and thin cloud pixels in the Landsat 7 dataset.

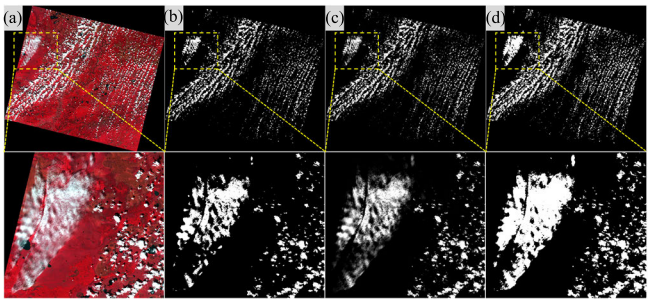


Fig. 6. Cloud boundary refinement with guided filtering. (a) Original image (green, red, and near-infrared band combination). (b) Thick cloud mask. (c) Guided filtering result. (d) Refined cloud mask.

D. Postprocessing

1) Morphological Opening Operation: Since the algorithm proposed in this article is not aimed at the pixel level instead of the superpixel object with large area, it is easy to introduce small-speckle noise after guided filtering. Specifically, it is often associated with noncloud pixels or scattered pixels that are not large enough to encompass a complete cloud object. Therefore, it is also necessary to remove such pixels. The morphological opening operation is a fast and effective method to eliminate small-speckle noise. It first carries out the erosion operation on the binary image, followed by a dilation operation, which not only retains the structural characteristics of the original shape but also effectively eliminates the isolated small-speckle noises. In this article, the morphological opening operation is carried out on the refined cloud mask to remove such noises while preserving the boundary of the whole cloud mask.

2) Hole Filling: There will inevitably be some holes in the cloud pixel area of the extracted cloud mask, which will lead to fewer detected cloud pixels than the actual number, resulting in omission error. Therefore, it is essential to fill holes to make the cloud mask more complete through counting the number of cloud pixels in the neighborhood of a certain window size around each noncloud pixel. If this number exceeds half of the window area, the noncloud pixel will be marked as a cloud pixel.

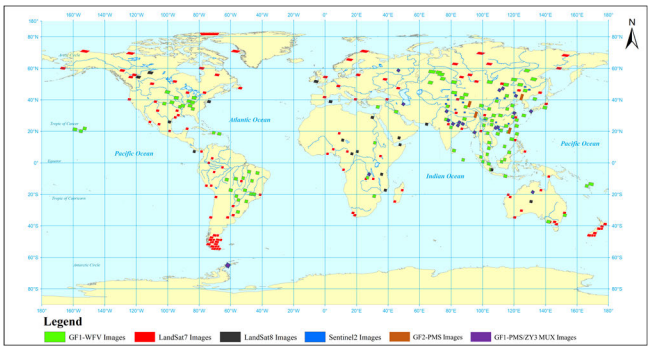


Fig. 7. Distribution of cloud detection images used in this article.

III. Dataset

To verify the results of the proposed algorithm on diverse images, we selected images from six public cloud detection datasets for experiments. These six datasets with distinct characteristics reflect the effect of cloud detection algorithms holistically. An overview of these datasets is given as follows.

The first GF1-WFV Cloud Detection dataset [10] is made up of 108 GaoFen-1 wide field view (WFV) Level-2A multispectral images acquired from May 2013 to August 2016 from different regions worldwide. The spatial resolution of the GF-1 WFV multispectral images is 16 m, and the spectral channels are BGR and NIR. The second dataset is Landsat7 Cloud Cover Assessment Validation dataset [37], which contains 206 precollection Landsat 7 Enhanced Thematic Mapper Plus (ETM+) Level-1 multispectral images with the resolution of 30 m. Similar to the Landsat7 dataset, the Landsat8_Collection dataset [38] has 48 images with the same resolution. The images were selected at random from the Landsat 8 archive all over the world. The corresponding cloud masks were interpolated as pixelwise clear, thin cloud, and thick cloud. For medium-resolution datasets, the WHU_S2_CD dataset [39] includes 36 Level-2A images and corresponding manually labeled cloud masks with GSD of 10 m from the Sentinel-2 satellite. These images are evenly distributed over China mainland. The WDCD dataset [29] was collected from GF-1 and ZiYuan-3 satellite images from all over the world. Its validation and testing datasets contain 30 images with pixel-level annotation labels with a spatial resolution from 5.8 to 8 m. As a high-resolution cloud detection dataset, the last AIR-CD dataset [25] contains 34 GF-2 satellite images and corresponding annotated masks with a spatial resolution of 4 m from different regions of China. These images were acquired from February 2017 to November 2017.

In this study, we selected 336 images from the above datasets after manual inspection and only extracted the BGR and NIR bands to form the new images to be detected. All of the images cover different landcover types, including urban areas, barren land, forest, sea, and snow mountains. These data have great differences in many aspects, such as region, sensor, spatial resolution, image size, and ground object type. The global distribution and details of the selected experimental images are shown in Fig. 7 and Table I, respectively.

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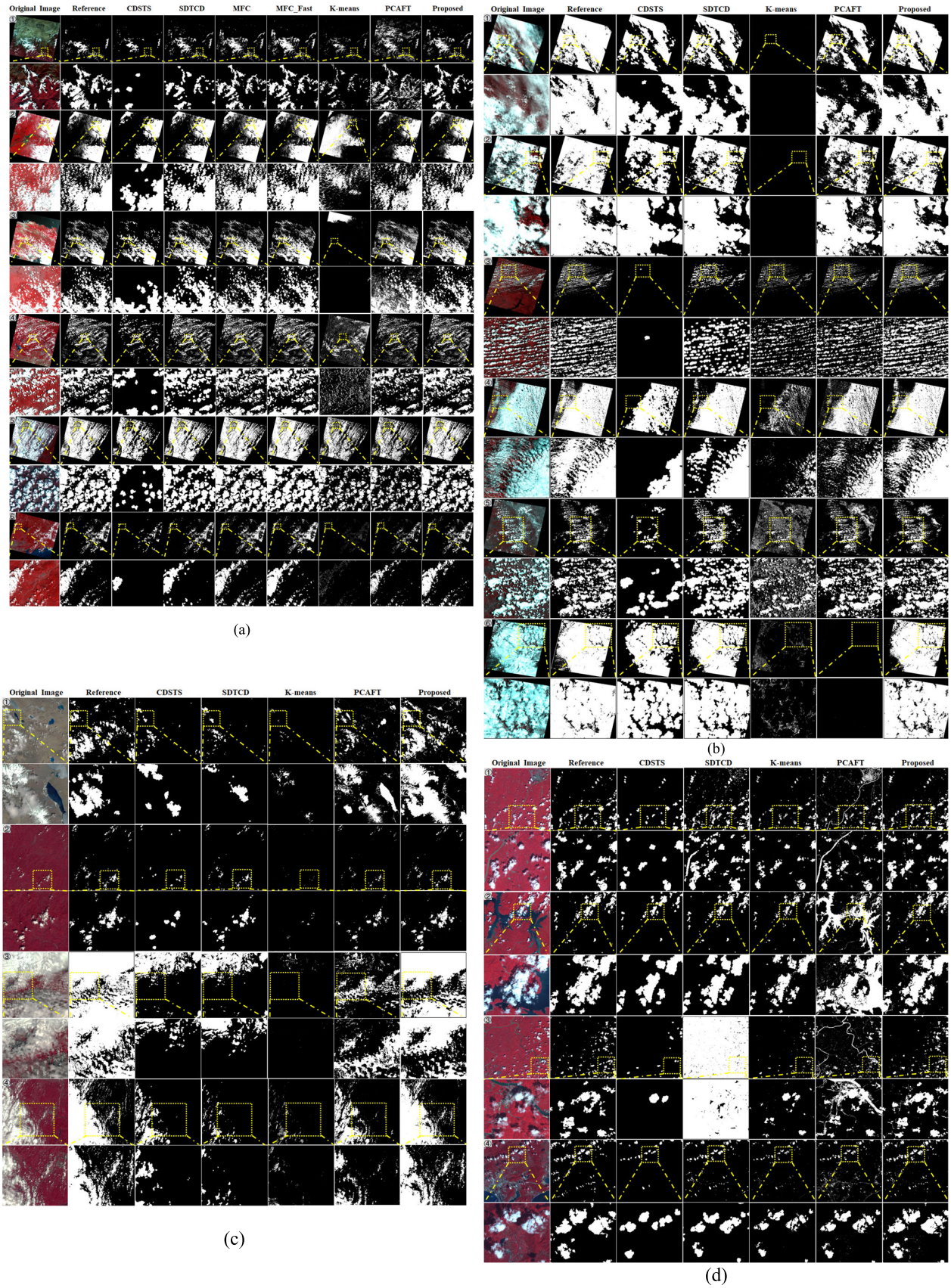


Fig. 8. Comparison of cloud detection results on the different cloud datasets. (a)–(f) Corresponding results of GF1-WFV, Landsat7, WHU_S2_CD, AIR_CD, WD_CD, and Landsat8_Collection1 datasets, respectively. From left to right, each column in turn shows the original remote sensing image (false-color composite image), the reference image, and the cloud detection results of the CDSTS algorithm, the SDTCD algorithm, the MFC/MFC_Fast algorithm (for GF1-WFV dataset only), K-means algorithm, PCAFT algorithm, and our proposed SMFCD algorithm. From top to bottom, each row shows cloud detection results from different images.

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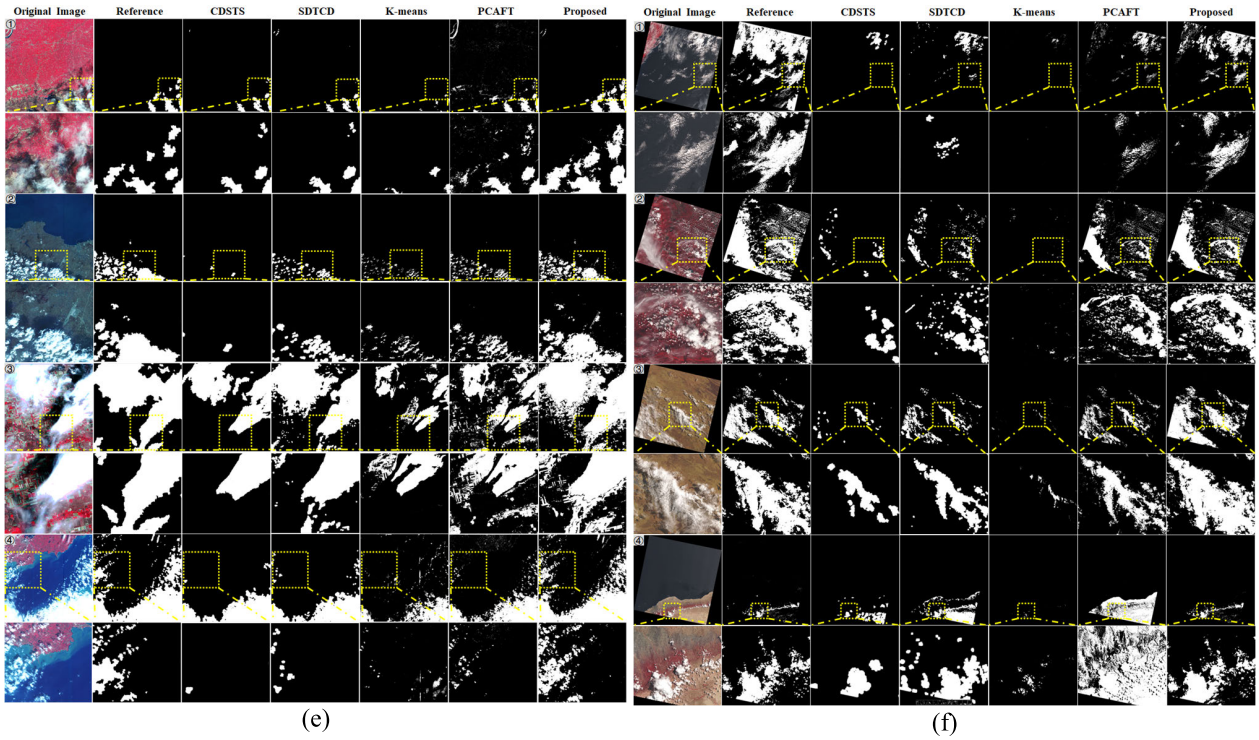


Fig. 8. *(Continued.)* Comparison of cloud detection results on the different cloud datasets. (a)–(f) Corresponding results of GF1-WFV, Landsat7, WHU_S2_CD, AIR_CD, WD_CD, and Landsat8_Collection1 datasets, respectively. From left to right, each column in turn shows the original remote sensing image (false-color composite image), the reference image, and the cloud detection results of the CDSTS algorithm, the SDTCD algorithm, the MFC/MFC_Fast algorithm (for GF1-WFV dataset only), K-means algorithm, PCAFT algorithm, and our proposed SMFCD algorithm. From top to bottom, each row shows cloud detection results from different images.

TABLE I
DETAILED INFORMATION OF EXPERIMENTAL SATELLITE IMAGES

Dataset	Satellite	Sensor	Type	Num	GSD/m	Size/pixel
GF1-WFV	GF-1	WFV	L2A	100+3	16	15701×16182
Landsat7	Landsat7	ETM+	L1	130+8	30	7320×8100
WHU_S2	Sential2	MSI	L1C	20+2	10	10980×10980
AIR_CD	GF-2	PMS1 PMS2	L1	19+3	4	6908×7300
WD_CD	GF1 ZY3	PMS MUX	L1	18+4	8 5.8	9283×8817 4544×4548
Landsat8_C1	Landsat8	OLI TIRS	L1	26+3	30	7852×7731

TABLE II
PARAMETER SETTINGS OF THE PROPOSED ALGORITHM

t_{seg}	t_{clear}	t_{area1}	t_{hwr1}	t_{area2}	t_{hwr2}	t_{ndwi1}	t_{ndwi2}	t_{GF_r}	t_{GF_t}
200	150	4000	3.4	200000	4.5	0.2	0.4	200	30

IV. EXPERIMENTAL RESULTS AND ANALYSIS

The parameters of the proposed algorithm are shown in Table II, which are selected after a large number of experiments.

The proposed algorithm is compared with the CDSTS algorithms [7], MFC algorithms [10], SDTCD [11] algorithms, K-means algorithm [40], and PCAFT algorithm [41]. The CDSTS algorithm used the superpixel segmentation and spectral threshold to generate the initial cloud mask; then, the texture features of the superpixels were extracted to eliminate the noncloud objects; and finally, the results were obtained after the region growth. Similarly, the SDTCD algorithm generates superpixels, the basic detection units, on the original image as well. Then, the spectral mean value of each superpixel is calculated, and the detection threshold is determined by the OTSU algorithm. Finally, the superpixel whose spectral mean is less than the threshold is regarded as noncloud pixel. The K-means cloud detection method is implemented based on the clustering algorithm. In this method, the cloud region is considered as the highest radiation homogeneous areas. By constructing anisotropic vectors composed of gray values of each pixel and its four neighbor pixels, k-means clustering is carried out to find the cloud pixels that are consistent with the centers of cloud clusters. Consistent with the central idea of the proposed algorithm, the PCAFT algorithm belongs to the saliency-based algorithm likewise. The frequency-tuned (FT) saliency [42] map of Gaussian filtering results at different scales is calculated, and the weighted fusion of the saliency results is carried out by using a principal component analysis (PCA). Finally, cloud extraction is carried out on the basis of the fusion results.

A. Cloud Detection Accuracy

We analyzed the results of applying the proposed algorithm and the above cloud detection algorithms to the six datasets from both qualitative and quantitative perspectives.

1) Qualitative Evaluation: Comparisons of the results of the various algorithms are shown in Fig. 8 (note that since the

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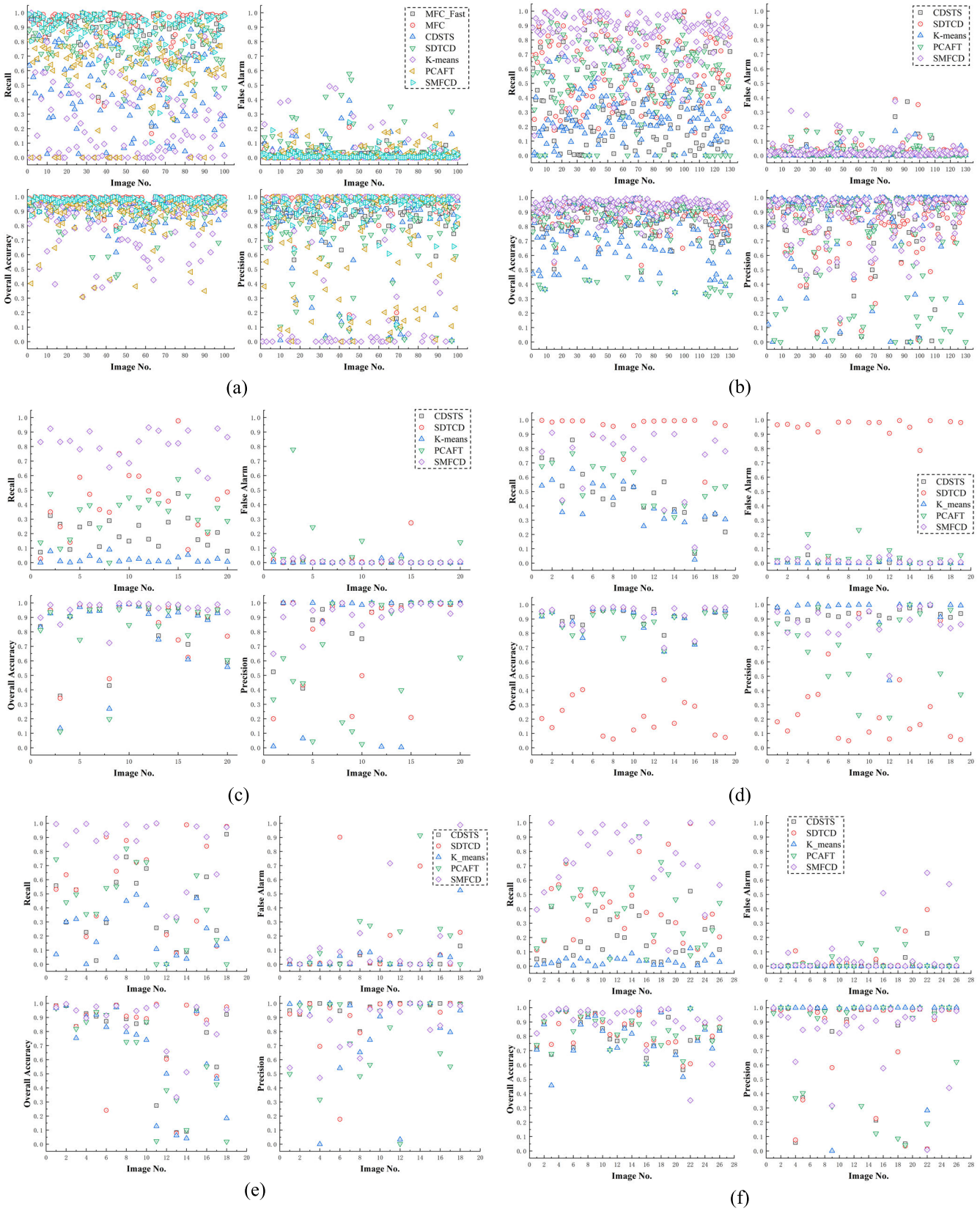


Fig. 9. Recall, FAR, OA, and precision of different algorithms on six cloud datasets. (a)–(f) Accuracies of GF1-WFV, Landsat7, WHU_S2_CD, AIR_CD, WD_CD, and Landsat8_Collection1 datasets, respectively.

MFC algorithm relies on the metadata of the GF1 imagery, it could not be applied to other datasets). According to the qualitative analysis, the proposed SMFCD algorithm gives significantly better results than other algorithms on all datasets,

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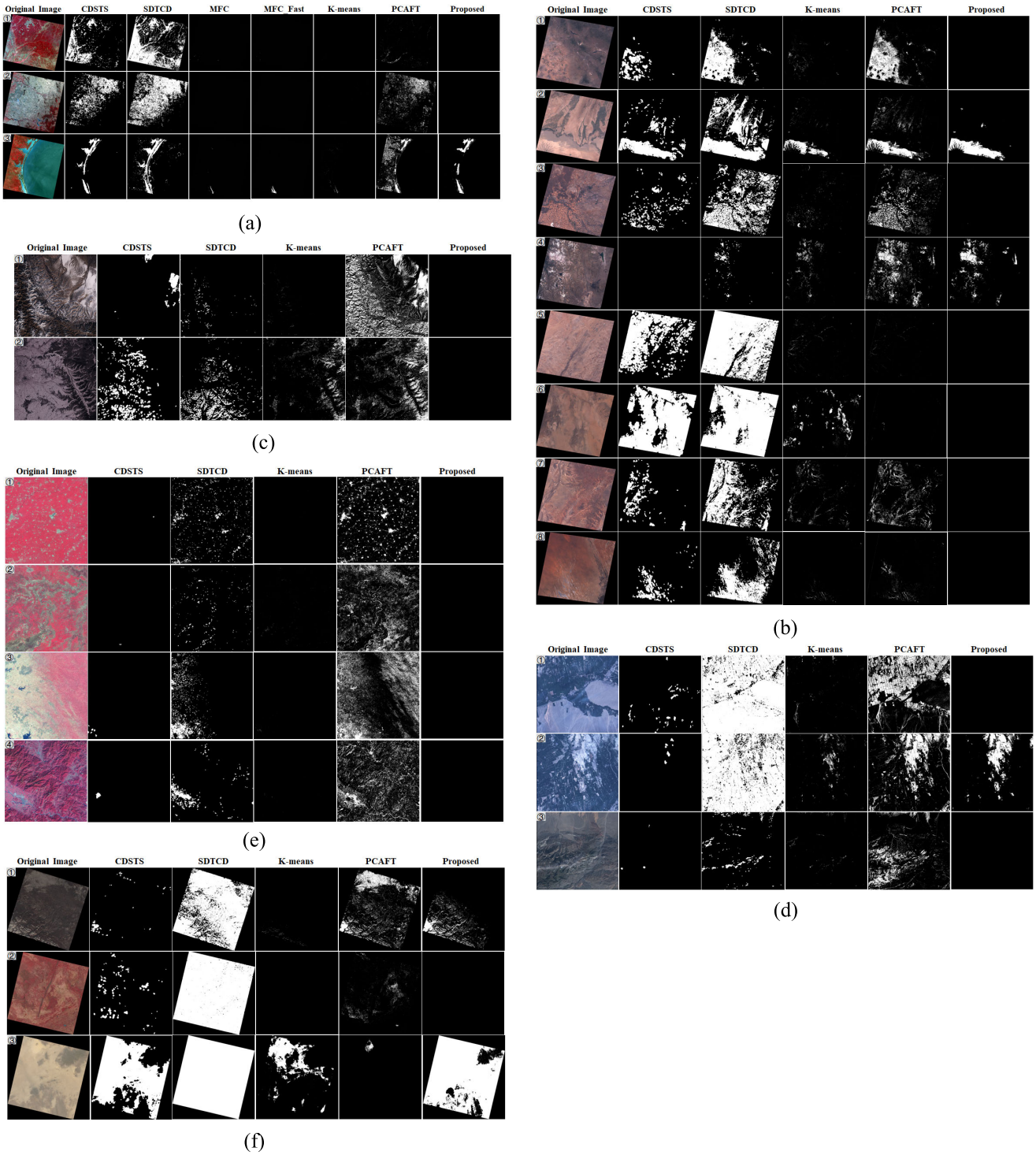


Fig. 10. Comparison of cloud detection results of clear images on six datasets. From left to right, each column in turn shows the original remote sensing image (false-color composite image) and the cloud detection results of the CDSTS algorithm, the SDTCD algorithm, the MFC/MFC_Fast algorithm (only for GF1-WFV dataset), the K-means algorithm, the PCAFT algorithm, and our proposed SMFCD algorithm. (a)–(f) Clear image detection results of GF1-WFV, Landsat7, WHU_S2_CD, AIR_CD,WD_CD, and Landsat8_Collection1 datasets, respectively.

with the derived cloud masks being more complete and including better edge detail. As can be seen from Fig. 8, on the GF1-WFV dataset, the result of the proposed SMFCD algorithm is close to that of the MFC algorithm. The CDSTS algorithm and the SDTCD algorithm can better extract obviously thick cloud clusters but have poor effect on thin clouds. The main reason is that both of these two methods rely on superpixel segmentation, which is hard to merge the thin and thick clouds together, resulting in the unsatisfactory effect of thin clouds. The K-means algorithm causes a large omission error on thin clouds as well and is prone to failure for large-area thick clouds [see Fig. 8(b)]. Another disadvantage is that noncloud pixels are easily mixed in the clustering process, especially those near thin clouds. The PCAFT algorithm is also

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TABLE III
MEAN VALUES OF RECALL, FAR, OA, AND PRECISION OF DIFFERENT ALGORITHMS ON THE SIX DATASETS

Dataset	Algorithm	Recall ↑	FAR ↓	OA ↑	Precision ↑
GF1-WFV	MFC	**0.897**	**0.008**	**0.978**	**0.953**
	MFC_Fast	0.879	0.025	0.966	0.866
	CDSTS	0.514	0.030	0.905	0.783
	SDTCD	0.771	0.068	0.904	0.723
	K-means	0.238	0.036	0.823	0.671
	PCAFT	0.582	0.032	0.869	0.675
	SMFCD	0.884	0.018	0.969	0.903
Landsat7	CDSTS	0.374	0.019	0.852	0.839
	SDTCD	0.595	0.032	0.883	0.816
	K-means	0.280	**0.008**	0.778	0.869
	PCAFT	0.537	0.024	0.820	0.732
	SMFCD	**0.817**	0.033	**0.933**	**0.870**
WHU_S2_CD	CDSTS	0.207	**0.002**	0.852	0.906
	SDTCD	0.416	0.019	0.855	0.802
	K-means	0.022	0.004	0.814	0.796
	PCAFT	0.326	0.075	0.816	0.643
	SMFCD	**0.811**	0.011	**0.955**	**0.922**
AIR_CD	CDSTS	0.473	**0.008**	0.916	0.937
	SDTCD	**0.944**	0.808	0.333	0.286
	K-means	0.406	0.002	0.897	**0.955**
	PCAFT	0.521	0.047	0.877	0.710
	SMFCD	0.714	0.020	**0.920**	0.865
WD_CD	CDSTS	0.413	**0.012**	0.743	**0.976**
	SDTCD	0.593	0.123	0.807	0.904
	K-means	0.212	0.054	0.641	0.797
	PCAFT	0.409	0.130	0.643	0.748
	SMFCD	**0.825**	0.142	**0.860**	0.845
LandSat8 Collection1	CDSTS	0.189	0.013	0.826	0.812
	SDTCD	0.409	0.035	0.832	0.788
	K-means	0.035	**0.000**	0.809	**0.933**
	PCAFT	0.389	0.034	0.831	0.735
	SMFCD	**0.756**	0.085	**0.894**	0.826

saliency-based, but it is inclined to blend water bodies into the detection results [see Fig. 8(d)], and its ability to exclude bright ground objects and detect thin clouds is inferior to that of our proposed SMFCD algorithm.

2) Quantitative Evaluation: To evaluate the accuracy of the derived cloud masks, the recall, false alarm rate (FAR), overall accuracy (OA), and precision are calculated as follows:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{18}$$

$$\text{False_Alarm} = \frac{\text{FP}}{\text{TN} + \text{FP}} \tag{19}$$

$$\text{Overall_Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Num Of Pixels}} \tag{20}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{21}$$

where TP, TN, FN, and FP are the numbers of true positives, true negatives, false negatives, and false positives, respectively, and NumOfPixels is the number of pixels in the image.

We quantitatively analyzed the cloud detection accuracy of all 336 images from different datasets. As can be seen from Fig. 9, the recall and precision of the CDSTS, SDTCD, K-means, and PCAFT algorithms are lower than those of the other algorithms. The distribution difference among all images is large, and the FAR is also significantly higher than those of the other algorithms. The recall, FAR, and precision of the proposed algorithm are obviously better than those of other algorithms on six different datasets and comparable to those of the MFC algorithm. In terms of OA, the proposed algorithm and the MFC algorithm are similar and are better than the other four algorithms. Fig. 9 also shows that our proposed algorithm has good accuracy consistency, indicating the ability to achieve good results in different scenarios and better robustness than the others.

As can be seen from Table III, in terms of all the average accuracy indices of the proposed algorithm when applied to the GF1-WFV dataset, it is better than other algorithms but slightly poorer than the MFC algorithm. The MFC algorithm is specifically designed for GF1-WFV imagery and has been optimized for the characteristics of this kind of data. Although it has higher accuracy, it is difficult to apply to other kinds of images at the same time. At this point, the proposed algorithm has a wider range of applicability. For the accuracies on the other five datasets, the average OA of the proposed algorithm is higher than those of other algorithms, illustrating that the proposed algorithm can acquire more reliable cloud detection results. As for recall, the proposed algorithm shows obvious advantages in the four datasets except for AID_CD, which is 19.3%–72.1% higher than other similar algorithms. Although the SDTCD algorithm has higher recall than the proposed algorithm on the AIR_CD dataset, it misidentifies most of the image pixels as clouds, which leads to the very high FAR and the lowest precision.

The above experimental results show that the proposed algorithm has good applicability to different types of remote sensing images. On the one hand, the proposed algorithm only requires the information of visible band and near-infrared band, which has stronger universality. On the other hand, compared with other similar algorithms, the OA of the proposed method on different datasets is among the best. The higher recall of the proposed method indicates that it has superior thin cloud identification ability than other algorithms.

3) Clear Image Recognition: Since clear images can seriously affect various quantitative indicators, the recognition results of each algorithm on clear images are compared separately. As can be seen from Fig. 10(a), for the GF1-WFV cloud dataset, the MFC algorithm and the K-means algorithm have the lowest cloud misjudgment rate for clear images. The proposed algorithm is significantly better than the other three algorithms in the recognition of clear images, but there are still some misjudgments in the area with bright water reflection [see Fig. 10(a)-③]. For the Landsat 7 dataset, the misjudgment rates of the CDSTS, SDTCD, and PCAFT algorithms are very high for highlighted ground objects. The proposed algorithm can correctly identify clear images except for snowy/icy scenes, as shown in Fig. 10(b)-④. It is worth noting that the CDSTS algorithm is accurate in discriminating regions with snow/ice on Landsat 7 images since the texture features extraction in this algorithm plays a key role in the recognition of cloud and snow. On the other four datasets, the proposed algorithm and the K-means algorithm have the best ability to recognize clear images except for high-light reflection scenes such as reflected water body [see Fig. 10(d)] and desert [see Fig. 10(f)].

B. Performance

The proposed cloud detection algorithm was implemented in C++ and run in parallel on a computer with an Intel i7-11700 CPU and 32-GB RAM. To further improve the

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TABLE IV
MEAN RUNNING TIMES OF DIFFERENT CLOUD DETECTION ALGORITHMS ON SIX DATASETS

Dataset	Algorithm	Time Cost(s)	Speedup
GF1-WFV	MFC	93.7	2.23
	MFC_Fast	24.2	0.57
	SDTCD	76.4	1.81
	CDSTS	47.4	1.13
	K-means	**753.0**	**17.89**
	PCAFT	69.2	1.64
	SMFCD	42.1	1.00
	SMFCD_Fast2	**19.5**	**0.46**
Landsat7	SDTCD	23.8	2.36
	CDSTS	10.5	1.04
	K-means	**85.3**	**8.45**
	PCAFT	14.8	1.47
	SMFCD	10.1	1.00
	SMFCD_Fast2	**5.7**	**0.56**
WHU_S2_CD	SDTCD	29.2	1.26
	CDSTS	24.9	1.07
	K-means	**298.5**	**12.87**
	PCAFT	34.0	1.47
	SMFCD	23.2	1.00
	SMFCD_Fast2	**14.2**	**0.61**
AIR_CD	SDTCD	**65.0**	**8.44**
	CDSTS	4.1	0.53
	K-means	62.6	8.13
	PCAFT	10.2	1.32
	SMFCD	7.7	1.00
	SMFCD_Fast2	**3.2**	**0.42**
WD_CD	SDTCD	6.2	1.41
	CDSTS	2.5	0.57
	K-means	**45.7**	**10.39**
	PCAFT	7.3	1.66
	SMFCD	4.4	1.00
	SMFCD_Fast2	**2.1**	**0.48**
Landsat8 Collection1	SDTCD	8.4	0.98
	CDSTS	6.8	0.79
	K-means	**109.3**	**12.71**
	PCAFT	13.8	1.60
	SMFCD	8.6	1.00
	SMFCD_Fast2	**4.9**	**0.57**

processing speed, we also propose a fast version of the proposed algorithm (termed SMFCD_Fast), where downsampling is first performed on the input original image according to a certain scale. The obtained cloud mask is then upsampled to the size of the original image eventually, thus accelerating the processing. The downsampling scale can be dynamically configured as a specified parameter.

We compare the performance of the proposed algorithm and its fast version with the other five algorithms that are under the same condition. From measurements of the cloud detection times of each algorithm on the six datasets, the average time consumption for single-image cloud detection is calculated. The speedup of average time between the proposed SMFCD algorithm and the others is computed for better efficiency comparison. As shown in Table IV, on the GF1-WFV dataset and WHU_S2_CD dataset with large image size, the proposed SMFCD_Fast2 algorithm (with a downsample scale of 2) has the shortest average time, while the K-means algorithm performs the longest. The SMFCD algorithm is much less than that of the MFC algorithm and the PCAFT algorithm. The SDTCD algorithm takes 2.4 and 8.4 times longer than the proposed SMFCD algorithm on the Landsat 7 dataset and AIR_CD dataset, respectively. The time of the CDSTS algorithm is close to that of the SMFCD algorithm on the WDCD dataset and the Landsat8_Collection1 dataset but slower than the fast version of the latter. On these two datasets, the SMFCD algorithm runs less than 5 and 10 s, and the corresponding fast version time is shortened by half. From this performance comparison, it can be seen that the proposed algorithm can achieve the shortest processing time on six datasets of different image sizes, which is of great significance in practical applications such as satellite image data quality detection system. The average accuracies and running time of each algorithm on different datasets are shown in Fig.11, so as to intuitively compare the performance of each algorithm.

TABLE V
MEAN RUNNING TIMES OF DIFFERENT CLOUD DETECTION ALGORITHMS ON DIFFERENT DATASETS

Dataset	Ratio combination	Recall ↑	FAR ↓	OA ↑	Precision ↑
GF1-WFV	HOT only	0.856	0.029	0.959	0.882
	0.25*HOT+0.75*BGR_Mean	0.876	0.020	0.968	0.900
	0.5*HOT+0.5*BGR_Mean	**0.884**	0.018	**0.969**	**0.903**
	0.75*HOT+0.25*BGR_Mean	0.880	0.023	0.964	0.880
	BGR_Mean only	0.837	**0.015**	0.962	0.905
Landsat7	HOT only	0.719	0.048	0.918	0.843
	0.25*HOT+0.75*BGR_Mean	0.792	0.035	0.931	**0.874**
	0.5*HOT+0.5*BGR_Mean	**0.817**	0.033	**0.933**	0.870
	0.75*HOT+0.25*BGR_Mean	0.804	0.032	0.931	0.868
	BGR_Mean only	0.766	**0.026**	0.924	0.884

C. Parameters Selection Analysis

1) Ratio Between HOT and Band Mean Map: For the mixed feature map, we used the weighted combination of HOT feature and BGR band mean feature. How to determine the respective weights is a key problem. In order to more intuitively show that the current 1:1 weight of the two features is optimal, we conducted experiments on two large datasets of GF1-WFV and Landsat7 under different weight combinations. As can be seen from Table V, the accuracy reaches the highest when the contribution of the HOT and BGR band mean is the same on both two datasets. Accordingly, we set the two weights to 0.5 in the calculation of the mixed feature map.

2) Guided Filter in Cloud Mask Refinement: In the cloud mask refinement stage, the two parameters $\boldsymbol{t}_{\mathrm{GF_}r}$ and $\boldsymbol{t}_{\mathrm{GF_}t}$ related to the guide filter also have a crucial impact on the accuracy. In the previous content, the recommended settings of $\boldsymbol{t}_{\mathrm{GF_}r}$ and $\boldsymbol{t}_{\mathrm{GF_}t}$ are obtained through a large number of experiments. We carried out cloud detection and OA calculation on the GF1-WFV dataset under different combinations of $\boldsymbol{t}_{\mathrm{GF_}r}$ and $\boldsymbol{t}_{\mathrm{GF_}t}$ and plotted the corresponding accuracy statistics into a 3-D figure in order to find the optimal combination intuitively. As shown in Fig. 12, the OA reaches the highest when $\boldsymbol{t}_{\mathrm{GF_}r}$ and $\boldsymbol{t}_{\mathrm{GF_}t}$ are set to 200 and 30, respectively (i.e., at the yellow five-pointed star).

3) Optimal Scale for Fast Version of SMFCD: We also analyzed the processing time and accuracy for different sampling scales. As can be seen in Fig. 13, when the sampling scale is less than 3, the running time is significantly reduced. At the same time, increasing the sampling scale does not significantly improve the performance. In addition, the higher the sampling scale, the lower the recall and the higher the FAR, but changes

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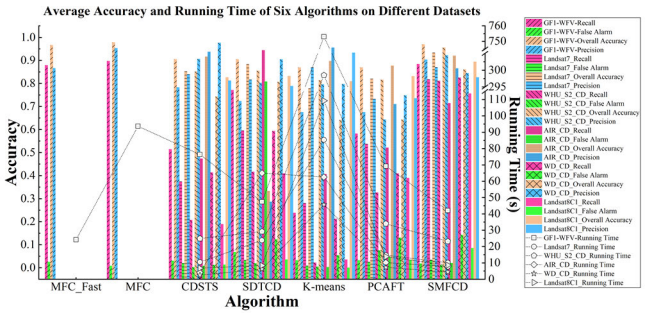


Fig. 11. Average accuracy and time consumption of different algorithms on six datasets.

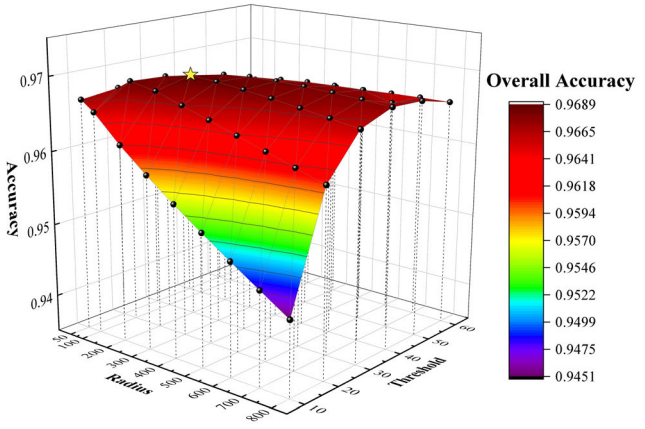


Fig. 12. Overall accuracy under different t_{GF_r} and t_{GF_t} parameters on the GF1-WFV dataset.

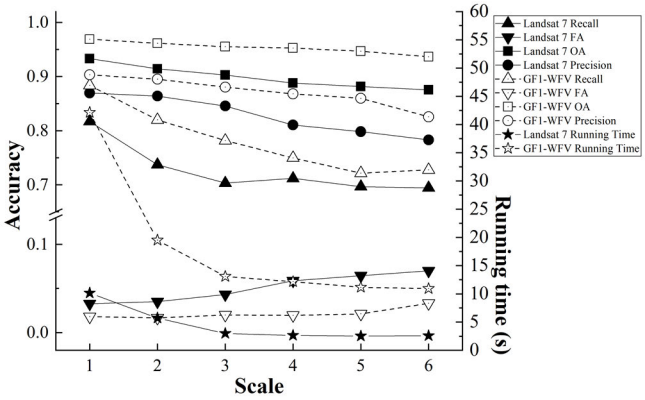


Fig. 13. Recall, FAR, OA, precision, and average time consumption for different sampling scales on GF1-WFV dataset and Landsat 7 dataset.

in sampling scale have relatively little influence on the overall precision and accuracy. Fig. 13 also shows that the optimal sampling scale for the algorithm proposed in this article is 2.

V. Conclusion

A remote sensing image cloud detection algorithm based on image saliency detection has been proposed. The algorithm directly takes the DN value of the original image as the input without the need to calculate the TOA reflectance. Therefore, the proposed algorithm does not depend on the radiometric calibration coefficients and is thus applicable to multispectral images from different satellites, giving it greater flexibility than existing algorithms. In addition, it requires fewer parameters, and the obtained cloud mask has better edge detail. Compared with other cloud detection algorithms, the proposed algorithm has advantages in terms of cloud detection accuracy and processing time. The OA of the proposed algorithm on six datasets can reach 0.860–0.969, which is 2.6%–34.2% higher than that of K-means algorithm. The recall can reach 0.756–0.884, and the precision is between 0.826 and 0.922. In terms of efficiency, for large-sized images (GF1-WFV or WHU_S2_CD), the proposed algorithm takes less than 20 s to detect cloud on average and less than 10 s on other datasets. The time of the proposed algorithm is only 44.9% of that of the MFC algorithm on the GF1-WFV dataset, while on other datasets, the K-means algorithm is 8.13–17.89 times slower, the SDTCD algorithm is 1.26–8.44 times slower, and the PCAFT algorithm is 1.32–1.66 times slower, compared with the proposed algorithm. All of the results and comparisons confirm that our proposed SMFCD algorithm can quickly and accurately extract cloud pixels in remote sensing images, thereby facilitating subsequent remote sensing image quality evaluation and other remote sensing applications.

At present, only four bands (three visible and NIR) are used in the proposed algorithm, and the ability to distinguish between cloud and snow cover needs to be further enhanced. In our future work, we will study cloud detection methods based on multisource data fusion. Most of the current cloud detection methods only use the band information of remote sensing images as input and ignore the geographic information of remote sensing images (such as latitude, longitude, and terrain elevation range), which is difficult to fully reflect all the information of the observation area. Therefore, the fusion of other auxiliary information and spectral information of bands can improve the detection performance and accuracy of the cloud detection algorithm. For example, by matching remote sensing images with DEM data corresponding to latitude and longitude, the mixed pixels of cloud and snow/ice in high mountain areas can be distinguished and eliminated to further improve the detection accuracy and discrimination ability of our proposed algorithm to such scenes.

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